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Project Title:	Experimental and Analytical Assessments of Polyethylene-Fiber Based Composites as Space-Radiation Shielding and Structural Materials		
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Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
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Zip Code:	35812	**Congressional District:**	5
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Task Description:	(1) Perform all necessary theoretical and accelerator-based analyses of the radiation-shielding effectiveness, including fatigue, of these materials. (2) Fabricate optimized material design panels for radiation exposure and generate database, summarizing results of the study on the effectiveness of the various combinations of material components on shielding efficiency. Develop and characterize multi-functional materials that can serve as integral structural members of the space vehicle while providing the necessary radiation shielding for the crew and are potentially, both mission enabling as well as cost effective. Identification and characterization of multifunctional materials		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:

Task Progress:

The objective of the proposed research effort was to evaluate the effectiveness of polyethylene (PE) based composites as multifunctional radiation shielding structural material through simulations and accelerator testing. The materials used in the study were graphite and PE reinforcing fibers, PE and epoxy resin matrices and metal hydrides. The addition of polyethylene fibers to the epoxy or polyethylene matrix composite added to the overall Hydrogen content of these materials, increasing their radiation shielding effectiveness as well as their mechanical properties.
The processing and fabrication of test panels for this study were conducted at the NASA Marshall Space Flight Center's National Center for Advanced Manufacturing (NCAM) facility. The thicknesses of the test panels for mechanical testing were chosen on the basis of American Society for Testing and Materials (ASTM) standards for mechanical testing and while those for radiation testing were chosen to yield statistically significant data during the limited beam time available. A hand lay-up processing technique was used to fabricate all test panels/articles. The mechanical and physical testing of the composite was performed at the NCAM facility. The test results show that these composites are well suited to be used as structural material.

Radiation Transport Calculations and Radiation Testing

The first phase of accelerator testing was performed at HIMAC. The purpose of this testing was to compare the transport calculations with the accelerator test data. NASA Space Radiation Shielding Project (SRSP) office had developed the material testing guide lines requiring the comparison of the new materials developed with the same arial density base line polyethylene. The second phase of the accelerator testing was performed at NASA Space Radiation Laboratory (NSRL) and SRSP guide lines were followed.

Phase I Transport Calculations

The calculations presented here are meant only to benchmark the radiation shielding effectiveness of the proposed PE-based composites relative to that for pure PE. In these simulations, we assume the radiation field is made up of GCR ions from protons (Z=1) through nickel (Z=28), ranging in kinetic energy from 1 MeV to 1 TeV per nucleon. Solar minimum conditions, where at maximum GCR flux, are assumed. GCR fluxes as functions of energy and relative abundances are taken to be those of nominal GCR spectra [1] at 1 AU appropriate for radiation-shielding simulations. To simulate a worst-case scenario, an unusually intense SEP field (the one associated with the October 1989 solar particle event) is added to the GCR field even though this addition amounts to about 7% of the total fluence for the simulated fluxes, energies, and material depths considered here. GCR ions are fully stripped, whereas heavy SEP ions are only partially stripped.

Comparison to theory was slightly complicated by the large-acceptance character of the detection system used. However, a preliminary simulation where all particles are assumed to be well resolved. A qualitative comparison between simulation and data is useful and can still be made for large Z fragments since these fragments contribute the most to the ionizing effects of high-energy and high-charge particulate radiation. Beam Exposure Data Obtained at NSRL

Other designs that are at various stages of development were also evaluated for radiation shielding effectiveness in addition to their comprehensive mechanical properties evaluation. Optimized material design specimens were fabricated using four different material combinations listed below:

a. 60% by volume of polyethylene fiber and 40% by volume of epoxy resin matrix composite of arial density 4.63 gm/cm2. (645)

b. 80% by volume polyethylene fiber and 20% by volume of polyethylene resin matrix composite of arial density 3.76 gm/cm2. (824)

c. 40% by volume of polyethylene fiber, 30% by volume of carbon fibers, and 30% by volume of epoxy resin matrix composite of arial density 3.28 gm/cm2. (G3C17)

d. PE/epoxy composite and Boron arial density 4.4 gm/cm2. (fc1)

Four specimens of the pure polyethylene (PE), of same aerial densities as above specimens were also fabricated as the base line material for comparison. All the specimens were exposed to 1000 MeV/nucleon Iron beam at NSRL. The test results indicate that the radiation shielding performance of equal arial density composites and PE is comparable. In addition the composites have superior mechanical properties compared to pure polyethylene, hence can be used as structural materials, leading to the multifunctional performance. The NSRL data was further reduced to calculate the track average LET and absorbed dose on the four different material combinations reported in last progress report for1000 MeV/nucleon Iron.

Phase II Transport Calculations

Radiation transport calculations were also performed for Polybenzoxazine resins (BZ3) that have high Hydrogen content. The results were compared with the epoxy resin (RS1) currently used in 645 composite. The results show enhanced radiation shielding capability of the BZ3 resin compared to the currently used RS1. The BZ3 resin was also compared to other high hydrogen epoxies. The BZ3 resin shows promise as radiation a shielding material. This resin will need to go through mechanical and physical testing to qualify it as multifunctional material. From all the test results so far it is concluded that candidate composite materials performed close to the base line polyethylene for radiation shielding. The mechanical and physical test results showed the expected superior performance compared to polyethylene.

Discussion The simulations and accelerator-based test results of the shielding effectiveness of an advanced composite design based on PE fiber in an epoxy matrix resin are presented. This study was part of the efforts at NASA-MSFC to design, fabricate and test multifunctional materials for use in deep-space mission planning and designs. The high hydrogen content of these composites gives them superior shielding effectiveness against high-energy and high-charge particle radiation fields that compose the deep-space radiation environment. Accelerator data suggest that this new design is able to attenuate the primary beam by 50% in about 5 cm. In addition, fluence data are able to depict the efficiency and degree to which this composite can reduce the expected radiation dose by primary-ion fragmentation and spallation. Coupled with high specific strength and modulus, this PE-based composite promises multifunctionality for deep-space exploration.

	References: [1] G.D. Badhwar and P.M. O'Neil, "Galactic Cosmic Radiation Model and Its Application", Adv. Space Res. 17, 1 (1996).
Bibliography Type:	Description: (Last Updated: 02/27/2004)